

Deep Generative Models

Lecture 10

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Recap of Previous Lecture

Continuous-Time Dynamics

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0.$$

$$\psi_t(\mathbf{x}_0) = \int_{t_0}^t \mathbf{v}_{\theta}(\mathbf{x}(s), s) ds + \mathbf{x}_0 \approx \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0, t).$$

- ▶ $\mathbf{v}_{\theta} : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ is a vector field.
- ▶ $\psi : \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R}^m$ is a flow (the solution of ODE):

Euler ODESolve

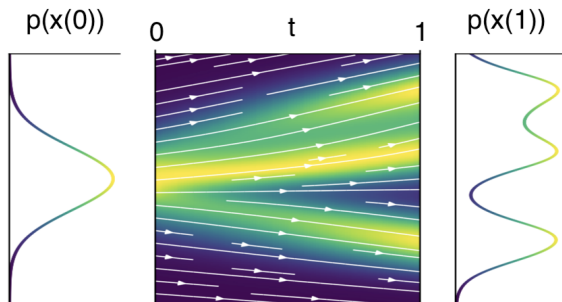
$$\mathbf{x}(t + h) = \mathbf{x}(t) + h \cdot \mathbf{v}_{\theta}(\mathbf{x}(t), t)$$

More advanced numerical methods (such as Runge-Kutta) are often used instead of unstable Euler update step.

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_\theta(\mathbf{x}(t), t); \quad \mathbf{x}(t_0) = \mathbf{x}_0$$

- ▶ Suppose $\mathbf{x}(0)$ is a random variable with density $p_0(\mathbf{x})$. Then, $\mathbf{x}(t)$ is a random variable with density $p_t(\mathbf{x})$.
- ▶ $p_t(\mathbf{x}) = p(\mathbf{x}, t)$ describes the **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.



Recap of Previous Lecture

Theorem (Picard)

If $\mathbf{v}$ is continuously differentiable with a bounded derivative in $\mathbf{x}$ and continuous in t , then the ODE has a **unique solution** given by a flow ψ_t .

This guarantees the ODE is **uniquely reversible**.

$$\psi_1(\mathbf{x}_0) = \mathbf{x}_0 + \int_0^1 \mathbf{v}_\theta(\psi_t(\mathbf{x}_0), t) dt$$

$$\mathbf{x}(1) = \mathbf{x}(0) + \int_0^1 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

$$\mathbf{x}(0) = \mathbf{x}(1) + \int_1^0 \mathbf{v}_\theta(\mathbf{x}(t), t) dt$$

Note: Unlike discrete-time flows, $\mathbf{v}$ need not be invertible (uniqueness ensures bijection).

How can we compute $p_t(\mathbf{x})$ at arbitrary t ?

Recap of Previous Lecture

$$\frac{d\mathbf{x}(t)}{dt} = \mathbf{v}_{\theta}(\mathbf{x}(t), t); \quad \text{with initial condition } \mathbf{x}(t_0) = \mathbf{x}_0$$

Theorem (Continuity Equation)

If $\mathbf{v}$ is continuously differentiable in $\mathbf{x}$ (with bounded derivative) and continuous in t , and $p_t(\mathbf{x}) > 0$, then

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

Solution

$$\log p_1(\mathbf{x}(1)) = \log p_0(\mathbf{x}(0)) - \int_0^1 \text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt.$$

- ▶ This solution gives us the density **along the trajectory**.
- ▶ However, it's difficult to efficiently estimate **the last term**.

Recap of Previous Lecture

SDE Basics

Let's define a stochastic process $\mathbf{x}(t)$ with initial condition $\mathbf{x}(0) \sim p_0(\mathbf{x})$:

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w},$$

where $\mathbf{w}(t)$ is the standard Wiener process (Brownian motion):

$$\mathbf{w}(t) - \mathbf{w}(s) \sim \mathcal{N}(0, (t - s)\mathbf{I}), \quad d\mathbf{w} = \epsilon \cdot \sqrt{dt}, \text{ where } \epsilon \sim \mathcal{N}(0, \mathbf{I}).$$

Discretizing the SDE (Euler-Maruyama Update) – SDEsolve

$$\mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{f}(\mathbf{x}(t), t) \cdot dt + g(t) \cdot \epsilon \cdot \sqrt{dt}$$

- ▶ At each time t , we have the density $p_t(\mathbf{x}) = p(\mathbf{x}, t)$.
- ▶ $p : \mathbb{R}^m \times [0, 1] \rightarrow \mathbb{R}_+$ is a **probability path** between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.

Recap of Previous Lecture

Theorem (Kolmogorov-Fokker-Planck)

If $p_t(\mathbf{x}) \in C^{1,2}$ (i.e., C^1 in t and C^2 in $\mathbf{x}$), then

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Langevin SDE (Special Case)

$$d\mathbf{x} = \frac{1}{2} \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) dt + \mathbf{1} \cdot d\mathbf{w}$$

The density $p_t(\mathbf{x})$ is a **stationary** distribution for the SDE. Setting the stationary density to the model distribution, $p_t(\mathbf{x}) = p_{\theta}(\mathbf{x})$ for all t :

Langevin Dynamic

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}} \log p_{\theta}(\mathbf{x}) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}, \quad \eta \approx dt.$$

Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)

Outline

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Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t,$$

$$q(\mathbf{x}_t|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

$$\mathbf{x}_{t-1} = \mathbf{x} + \sigma_{t-1} \cdot \boldsymbol{\epsilon}_{t-1},$$

$$q(\mathbf{x}_{t-1}|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_{t-1}^2 \cdot \mathbf{I})$$

Score Matching SDE

Denoising Score Matching

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$$\mathbf{x}_t = \mathbf{x}_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \cdot \boldsymbol{\epsilon}, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_{t-1}, (\sigma_t^2 - \sigma_{t-1}^2) \cdot \mathbf{I})$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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Let's transform this Markov chain into the continuous stochastic process $\mathbf{x}(t)$ by letting $T \rightarrow \infty$:

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \epsilon$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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$$\begin{aligned} \mathbf{x}(t) &= \mathbf{x}(t - dt) + \sqrt{\sigma^2(t) - \sigma^2(t - dt)} \cdot \boldsymbol{\epsilon} \\ &= \mathbf{x}(t - dt) + \sqrt{\frac{\sigma^2(t) - \sigma^2(t - dt)}{dt}} dt \cdot \boldsymbol{\epsilon} \end{aligned}$$

Score Matching SDE

Denoising Score Matching

$$\mathbf{x}_t = \mathbf{x} + \sigma_t \cdot \boldsymbol{\epsilon}_t, \quad q(\mathbf{x}_t | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma_t^2 \cdot \mathbf{I})$$

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Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
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Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$ is a monotonically increasing function.

Score Matching SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) + \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$
$$d\mathbf{x} = \mathbf{x}(t) - \mathbf{x}(t - dt)$$

Variance Exploding SDE

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

$\sigma(t)$ is a monotonically increasing function.

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = 0, \quad g(t) = \sqrt{\frac{d[\sigma^2(t)]}{dt}}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

Let's turn this Markov chain into a continuous stochastic process by letting $T \rightarrow \infty$ and setting $\beta_t = \beta(\frac{t}{T}) \cdot \frac{1}{T}$ (where $dt = \frac{1}{T}$):

$$\mathbf{x}(t) = \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

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$$\begin{aligned} \mathbf{x}(t) &= \sqrt{1 - \beta(t)dt} \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \approx \\ &\approx \left(1 - \frac{1}{2}\beta(t)dt\right) \cdot \mathbf{x}(t - dt) + \sqrt{\beta(t)dt} \cdot \epsilon \end{aligned}$$

Diffusion SDE

Denoising Diffusion

$$\mathbf{x}_t = \sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t, \quad q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} \cdot \mathbf{x}_{t-1}, \beta_t \cdot \mathbf{I})$$

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Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

Diffusion SDE

$$\mathbf{x}(t) = \mathbf{x}(t - dt) - \frac{1}{2}\beta(t)\mathbf{x}(t - dt)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

$$\mathbf{f}(\mathbf{x}, t) = -\frac{1}{2}\beta(t)\mathbf{x}(t), \quad g(t) = \sqrt{\beta(t)}$$

Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Diffusion SDE

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

Variance Exploding SDE (NCSN)

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w}$$

Variance Preserving SDE (DDPM)

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

We treat the discrete-time methods (NCSN, DDPM) as a special case of the continuous-time methods (VE-SDE, VP-SDE).

Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
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Probability Flow ODE

ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

$$\frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) \Leftrightarrow \frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x}))$$

The only source of randomness is the initial distribution $p_0(\mathbf{x})$.

Probability Flow ODE

ODE and Continuity Equation

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt$$

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The only source of randomness is the initial distribution $p_0(\mathbf{x})$.

SDE and KFP Equation

$$d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}g^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

Now there are two sources of randomness: the initial distribution $p_0(\mathbf{x})$ and the Wiener process $\mathbf{w}(t)$.

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

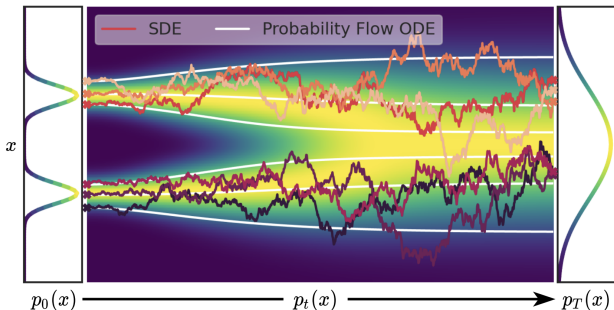
$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Probability Flow ODE

Theorem

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Proof

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right)$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

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Proof

$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) \end{aligned}$$

Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

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Probability Flow ODE

Theorem

Suppose the SDE $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ induces the probability path $p_t(\mathbf{x})$ with $p_t(\mathbf{x}) > 0$. Then, there exists an ODE with the same probability path $p_t(\mathbf{x})$, given by

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

Proof

$$\begin{aligned} \frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x})] + \frac{1}{2}g^2(t)\frac{\partial^2 p_t(\mathbf{x})}{\partial \mathbf{x}^2} \right) = \\ &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)\frac{\partial p_t(\mathbf{x})}{\partial \mathbf{x}} \right] \right) = \\ &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\mathbf{f}(\mathbf{x}, t)p_t(\mathbf{x}) - \frac{1}{2}g^2(t)p_t(\mathbf{x})\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right] \right) = \\ &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right) \end{aligned}$$

Probability Flow ODE

Proof (Continued)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right)$$

Probability Flow ODE

Proof (Continued)

$$\begin{aligned}\frac{\partial p_t(\mathbf{x})}{\partial t} &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} \left[\left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2} g^2(t) \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) p_t(\mathbf{x}) \right] \right) = \\ &= \text{tr} \left(-\frac{\partial}{\partial \mathbf{x}} [\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x})] \right) = -\text{div}(\mathbf{v}(\mathbf{x}, t) p_t(\mathbf{x}))\end{aligned}$$

Probability Flow ODE

Proof (Continued)

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$$\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}); \quad \tilde{g}(t) = 0$$

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt + 0 \cdot d\mathbf{w} = \left(\mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x}) \right) dt$$

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) + \frac{1}{2}\tilde{g}^2(t)\Delta_{\mathbf{x}}p_t(\mathbf{x})$$

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- $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ is the continuous-time score function.

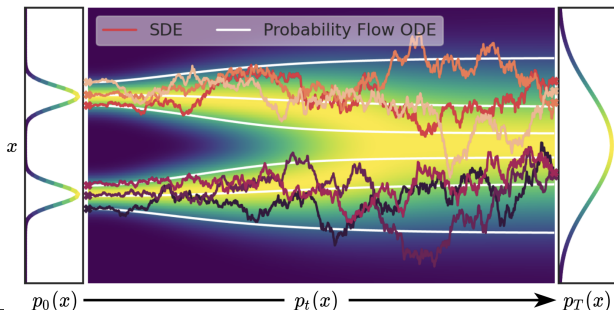
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- ▶ $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$ is the continuous-time score function.
- ▶ The ODE produces more stable trajectories.



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Why convert SDEs to PF-ODEs? (1) Efficient Sampling

- ▶ Brownian increments scale as $d\mathbf{w} \sim \sqrt{dt}$ — SDE trajectories are nowhere differentiable.
- ▶ SDEsolve (Euler–Maruyama): local error $\mathcal{O}(\sqrt{\Delta t})$.
- ▶ Higher-order SDE schemes need Lévy areas $\Rightarrow$ impractical in high dimensions.

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- ▶ Adaptive step-size control is standard for ODEsolve.
- ▶ In practice: 100–1000 SDEsolve steps $\rightarrow$ 20–50 ODEsolve steps.
- ▶ Caveat: SDE sampling can still win at high NFE — injected noise corrects accumulated error.

Why convert SDEs to PF-ODEs? (2) Deterministic Generation

- ▶ The PF-ODE defines a deterministic, invertible map $\mathbf{x}(0) \leftrightarrow \mathbf{x}(T)$.
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Exact log-likelihood (instantaneous change of variables)

$$\log p_0(\mathbf{x}(0)) = \log p_T(\mathbf{x}(T)) + \int_0^T \text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right) dt$$

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- ▶ Connects PF-ODE to continuous normalizing flows.
- ▶ Invertibility enables image $\rightarrow$ latent $\rightarrow$ edited image workflows.
- ▶ Smooth interpolation in latent space between two samples.

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Same trained score $\mathbf{s}_\theta(\mathbf{x}, t)$; only the discretization changes (reverse SDE derived in the next section).

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Sampling (Reverse SDE, Euler–Maruyama)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$.
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$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t - \left[\mathbf{f}(\mathbf{x}_t, t) - \frac{1}{2} g^2(t) \mathbf{s}_\theta(\mathbf{x}_t, t) \right] \Delta t \quad (\text{no noise injection})$$

Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)

Reverse SDE

$$d\mathbf{x} = \mathbf{v}(\mathbf{x}, t)dt, \quad \mathbf{x}(t + dt) = \mathbf{x}(t) + \mathbf{v}(\mathbf{x}, t)dt$$

Here dt can be > 0 or < 0 .

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Let $\tau = 1 - t$ ($d\tau = -dt$).

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There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

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where $dt < 0$ and $\bar{\mathbf{w}}$ is a reverse-time Wiener process.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

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- ▶ Here the score function appears: $\mathbf{s}(\mathbf{x}, t) = \frac{\partial}{\partial \mathbf{x}} \log p_t(\mathbf{x})$.
- ▶ Since $\mathbf{v}(\mathbf{x}, t) = \mathbf{f}(\mathbf{x}, t) - \frac{1}{2}g^2(t)\mathbf{s}(\mathbf{x}, t)$ (PF-ODE), the reverse drift equals $\mathbf{f} - g^2\mathbf{s} = \mathbf{v} - \frac{1}{2}g^2\mathbf{s}$.

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Proof Sketch

- ▶ Convert the initial SDE to a probability flow ODE.
- ▶ Reverse the probability flow ODE.
- ▶ Convert the reversed probability flow ODE back to an SDE.

Song Y., et al. Score-Based Generative Modeling through Stochastic Differential Equations, 2020

Proof of Reverse SDE

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Here $\mathbf{w}(\tau)$ is a Wiener process in forward τ -time.

Reverse SDE

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There exists a reverse SDE for $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$ (assume $p_t(\mathbf{x}) > 0$), given by:

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Here $d\tau > 0$, $dt < 0$, and $\bar{\mathbf{w}}$ is reverse-time Wiener process.

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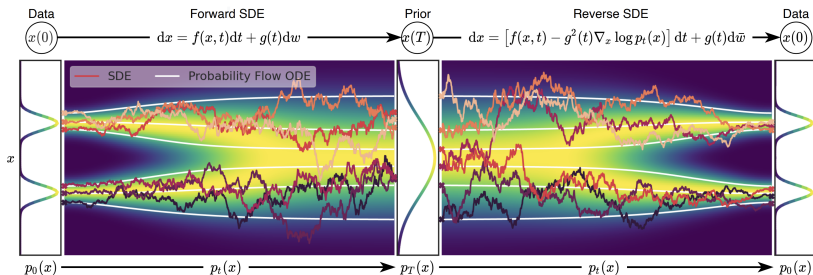
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Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Score Matching SDE

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$\sigma(t)$ is a monotonically increasing function.

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Diffusion SDE

Variance Preserving SDE

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w}$$

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Variance is preserved as long as $\mathbf{x}(0)$ has unit variance.

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Outline

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Score-Based Generative Models Through SDEs

Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

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Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$

Score-Based Generative Models Through SDEs

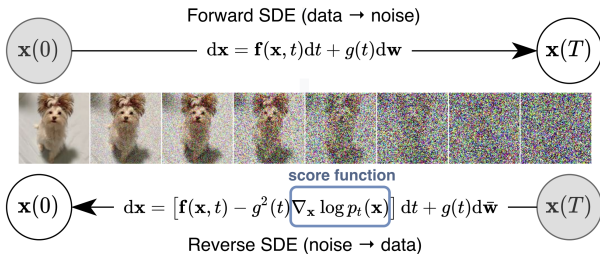
Discrete-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}_0)} \mathbb{E}_{t \sim U\{1, T\}} \mathbb{E}_{q(\mathbf{x}_t | \mathbf{x}_0)} \left\| \mathbf{s}_{\theta, t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t | \mathbf{x}_0) \right\|_2^2$$

Is it possible to train score-based diffusion models in continuous time?

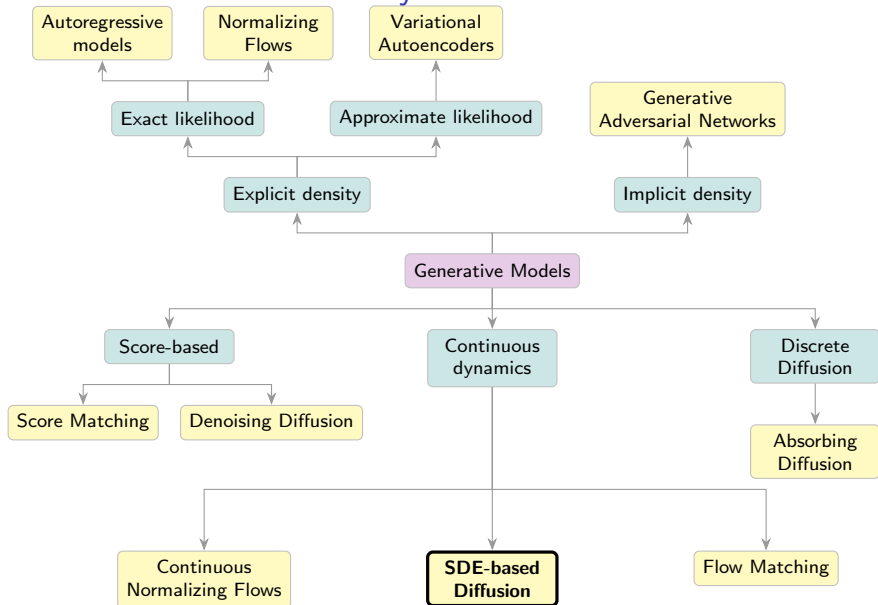
Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0, 1]} \mathbb{E}_{q(\mathbf{x}(t) | \mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t) | \mathbf{x}(0)) \right\|_2^2$$



Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Generative Models Taxonomy



Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\right)$$

$$\nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) = -\frac{1}{\sigma^2(t)}(\mathbf{x}(t) - \boldsymbol{\mu})$$

Note: Normality holds for $\mathbf{f}(\mathbf{x}, t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$).

Score-Based Generative Models Through SDEs

Continuous-Time Objective

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x}(0))} \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{q(\mathbf{x}(t)|\mathbf{x}(0))} \left\| \mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0)) \right\|_2^2$$

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Note: Normality holds for $\mathbf{f}(\mathbf{x}, t)$ affine in $\mathbf{x}$ ($\mathbf{f}(\mathbf{x}, t) = f(t)\mathbf{x}$).

$$d\mathbf{x} = \sqrt{\frac{d[\sigma^2(t)]}{dt}} \cdot d\mathbf{w} \quad (\text{Variance Exploding SDE})$$

$$d\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}(t)dt + \sqrt{\beta(t)} \cdot d\mathbf{w} \quad (\text{Variance Preserving SDE})$$

Is it possible to explicitly derive $\boldsymbol{\mu}(\mathbf{x}(0), t)$ and $\sigma^2(t)$ for VE-SDE and VP-SDE?

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\right)$$

Theorem

The moments of the SDE $d\mathbf{x} = f(t)\mathbf{x}dt + g(t)d\mathbf{w}$ satisfy:

$$\frac{d\boldsymbol{\mu}(\mathbf{x}(0), t)}{dt} = f(t)\boldsymbol{\mu}(\mathbf{x}(0), t)$$

$$\frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t)$$

Score-Based Generative Models Through SDEs

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$$\frac{d\sigma^2(t)}{dt} = 2f(t)\sigma^2(t) + g^2(t)$$

Proof

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)]$$

Score-Based Generative Models Through SDEs

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Proof

$$\begin{aligned}\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] &= \mathbb{E}[f(t)\mathbf{x}dt|\mathbf{x}(0)] + \mathbb{E}[g(t)d\mathbf{w}|\mathbf{x}(0)] \\ &= f(t)\mathbb{E}[\mathbf{x}|\mathbf{x}(0)] dt + g(t)\mathbb{E}[d\mathbf{w}|\mathbf{x}(0)]\end{aligned}$$

Score-Based Generative Models Through SDEs

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\boldsymbol{\mu}(\mathbf{x}(0), t), \sigma^2(t) \cdot \mathbf{I}\right)$$

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Proof

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Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$

Score-Based Generative Models Through SDEs

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Proof (Continued)

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = f(t)\mu(\mathbf{x}(0), t)dt$$

Score-Based Generative Models Through SDEs

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$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$

Proof (Continued)

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Score-Based Generative Models Through SDEs

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Examples

$$\textbf{NCSN: } f(t) = 0 \quad \Rightarrow \quad \mu = \mathbf{x}(0)$$

Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$

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$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = f(t)\mu(\mathbf{x}(0), t)dt$$

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$$\textbf{NCSN: } f(t) = 0 \quad \Rightarrow \quad \mu = \mathbf{x}(0)$$

$$\textbf{DDPM: } f(t) = -\frac{1}{2}\beta(t) \quad \Rightarrow \quad \frac{d\mu}{dt} = -\frac{1}{2}\beta(t)\mu$$

Score-Based Generative Models Through SDEs

Theorem

$$\frac{d\mu(\mathbf{x}(0), t)}{dt} = f(t)\mu(\mathbf{x}(0), t)$$

Proof (Continued)

$$\mathbb{E}[d\mathbf{x}|\mathbf{x}(0)] = f(t)\mu(\mathbf{x}(0), t)dt$$

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$$\textbf{NCSN: } f(t) = 0 \quad \Rightarrow \quad \mu = \mathbf{x}(0)$$

$$\textbf{DDPM: } f(t) = -\frac{1}{2}\beta(t) \quad \Rightarrow \quad \frac{d\mu}{dt} = -\frac{1}{2}\beta(t)\mu$$

$$\mu = \mathbf{x}(0) \exp\left(-\frac{1}{2} \int_0^t \beta(s) ds\right)$$

Score-Based Generative Models Through SDEs

Training

1. Sample $\mathbf{x}(0) \sim p_{\text{data}}(\mathbf{x})$, $t \sim U[0, 1]$.
2. Sample $\mathbf{x}(t) \sim q(\mathbf{x}(t)|\mathbf{x}(0))$.
3. Compute loss $\mathcal{L} = \|\mathbf{s}_{\theta}(\mathbf{x}(t), t) - \nabla_{\mathbf{x}(t)} \log q(\mathbf{x}(t)|\mathbf{x}(0))\|_2^2$.

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Score-Based Generative Models Through SDEs

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NCSN

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0), [\sigma^2(t) - \sigma^2(0)] \cdot \mathbf{I}\right)$$

Score-Based Generative Models Through SDEs

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NCSN

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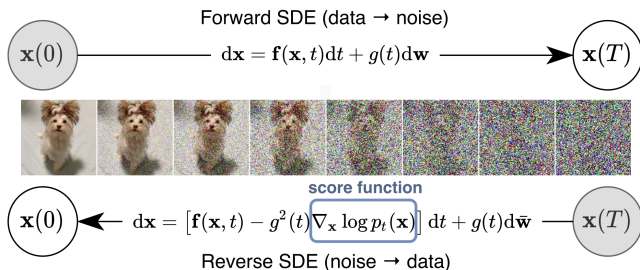
DDPM

$$q(\mathbf{x}(t)|\mathbf{x}(0)) = \mathcal{N}\left(\mathbf{x}(0)e^{-\frac{1}{2}\int_0^t \beta(s)ds}, \left(1 - e^{-\int_0^t \beta(s)ds}\right) \cdot \mathbf{I}\right)$$

Score-Based Generative Models Through SDEs

Sampling

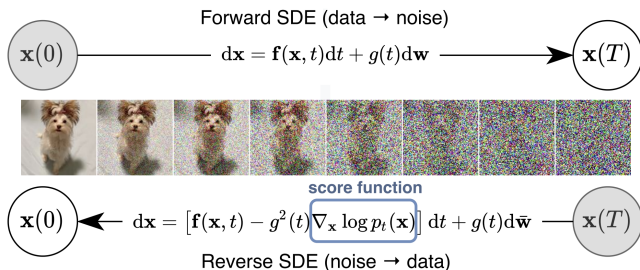
1. Sample $\mathbf{x}(1) \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the reverse SDE using numerical solvers (SDEsolve).



Score-Based Generative Models Through SDEs

Sampling

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- ▶ Discretizing the reverse SDE provides ancestral sampling.
- ▶ Discretizing the probability flow ODE yields deterministic sampling.

Song Y., et al. *Score-Based Generative Modeling through Stochastic Differential Equations*, 2020

Outline

1. Diffusion and Score Matching SDEs (VP-SDE / VE-SDE)
2. Probability Flow ODE
3. Reverse SDE
4. Score-Based Generative Models Through SDEs
5. Flow Matching (FM)

Continuous-Time Normalizing Flows

Let's return to ODE dynamics $\mathbf{x}_t = \mathbf{x}(t)$ in the interval $t \in [0, 1]$:

- ▶ $\mathbf{x}_0 \sim p_0(\mathbf{x})$, where $p_0(\mathbf{x})$ is a base distribution (mostly $\mathcal{N}(0, \mathbf{I})$);
- ▶ $\mathbf{x}_1 \sim p_1(\mathbf{x})$, where $p_1(\mathbf{x})$ is the true data distribution $p_{\text{data}}(\mathbf{x})$.

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}(\mathbf{x}_t, t), \quad \text{with initial condition } \mathbf{x}(0) = \mathbf{x}_0.$$

The velocity $\mathbf{v}$ determines a flow ψ_t : $\mathbf{x}_t = \psi_t(\mathbf{x}_0)$.

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KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\text{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\text{tr} \left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)} \right)$$

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- ▶ It's hard to solve the continuity equation directly due to the trace term.
- ▶ There's a method (the adjoint method) that solves this equation directly, but it's unstable and unscalable.

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

- ▶ Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- ▶ Flow matching provides scalable approach to Neural ODEs.

Continuous-Time Normalizing Flows

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- ▶ Knowing the vector field $\mathbf{v}(\mathbf{x}, t)$, the KFP (or continuity) equation allows us to compute the density $p_t(\mathbf{x})$.
- ▶ Flow matching provides scalable approach to Neural ODEs.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

Continuous-Time Normalizing Flows

KFP Theorem (Continuity Equation)

$$\frac{\partial p_t(\mathbf{x})}{\partial t} = -\operatorname{div}(\mathbf{v}(\mathbf{x}, t)p_t(\mathbf{x})) \Leftrightarrow \frac{d \log p_t(\mathbf{x}(t))}{dt} = -\operatorname{tr}\left(\frac{\partial \mathbf{v}(\mathbf{x}(t), t)}{\partial \mathbf{x}(t)}\right)$$

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- ▶ Approximate the true vector field $\mathbf{v}(\mathbf{x}, t)$ using $\mathbf{v}_\theta(\mathbf{x}, t)$.
- ▶ Use the learned flow ψ_t for deterministic sampling:
 $\mathbf{x}_1 = \psi_1(\mathbf{x}_0)$.

Flow Matching: Training and Sampling

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

$$\mathbf{x}_0 \sim p_0(\mathbf{x}) = p(\mathbf{x}), \quad \mathbf{x}_1 \sim p_1(\mathbf{x}) = p_{\text{data}}(\mathbf{x})$$

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Training

1. Sample $t \sim U[0, 1]$, $\mathbf{x}_t \sim p_t(\mathbf{x})$.
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Sampling

1. Sample $\mathbf{x}_0 \sim \mathcal{N}(0, \mathbf{I})$.
2. Solve the ODE to obtain $\mathbf{x}_1$:

$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Flow Matching: Training and Sampling

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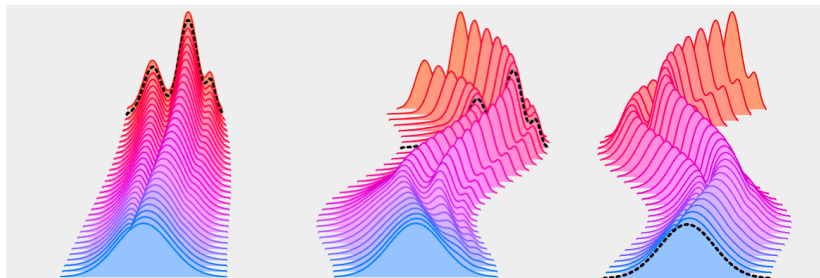
$$\mathbf{x}_1 = \psi_1(\mathbf{x}_0) = \text{ODESolve}_v(\mathbf{x}_0, \theta, t_0 = 0, t_1 = 1).$$

Problem: The true vector field $\mathbf{v}(\mathbf{x}, t)$ is **unknown**.

Flow Matching

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x})} \|\mathbf{v}(\mathbf{x}, t) - \mathbf{v}_\theta(\mathbf{x}, t)\|^2 \rightarrow \min_{\theta}$$

- ▶ There are infinitely many possible $\mathbf{v}(\mathbf{x}, t)$ between $p_0(\mathbf{x})$ and $p_1(\mathbf{x})$.
- ▶ We need to select the "best" $\mathbf{v}(\mathbf{x}, t)$ and make the objective tractable.



Summary

- ▶ Score matching (NCSN) and diffusion models (DDPM) are discretizations of SDEs (variance exploding and variance preserving).
- ▶ Every SDE admits a corresponding probability flow ODE that follows the same probability path $p_t(\mathbf{x})$, yielding deterministic invertible sampling and exact log-likelihoods via the instantaneous change of variables.
- ▶ Converting SDEs to PF-ODEs enables efficient high-order numerical solvers, reducing the number of inference steps from 100–1000 to 20–50.
- ▶ SDEs can be reversed in time using the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$; discretizing the reverse SDE gives ancestral sampling, while discretizing the PF-ODE gives deterministic sampling.
- ▶ The continuous-time score matching objective generalizes DDPM and NCSN; for affine-drift SDEs the transition kernel $q(\mathbf{x}(t)|\mathbf{x}(0))$ is Gaussian, making the training objective tractable.